NESTED-SPACE COSMOLOGY

One Recursive Spectral Gradient–Boundary Equation Across Scales

A constructive one-equation framework and its current reproducible evidence

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Contents

Abstract

Nested-Space Cosmology proposes that local matter, inherited constants, unresolved gravitational response, and a finite parent-to-child transition arise from one recursive operator and one set of laws. The organizing condition is $T_{\Theta}^{*}D_{\Theta}=D_{\Theta}$. The objective is a self-sourced physical solution whose matter, geometry and observations follow from the same action and state.

The reproducible foundation includes evaluated full-spinor boundary maps, normalized recursion, smooth geometric spectra, causal Dirac transport, quantum vacuum-source controls, and a scalar sheet coupling with an invariant massive Dirac sector. A prescribed geometry pulse creates Dirac pairs with a verified work balance. Covariant metric variation and normalized state histories identify further pieces of the common quantum functional.

This revision combines affine-horizon state data with the actual complex exterior Dirac reflection. The resulting canonical massless candidate has mean radial null stress approximately -0.05280 at the imposed neck, in units $\hbar=c=L_{throat}=1$ and normalization $\mu=1$. Both null contractions are negative after including the flux. No scalar source or Einstein coefficient was fitted. Density, pressure anisotropy and flux still leave independent geometric residuals.

A separate thermal calculation quantifies the finite state-dependent conversion required by the adopted proper-time prescription. The canonical state source cannot be identified directly with the raw finite determinant. The same-state nonthermal conversion and full domain conditions remain to be established before the complete source can be used for backreaction.

The 100-record collection preserves 58 historical records and 42 scoped follow-ups. Evaluated boundary, state and source connections form a reproducible path toward the proposed common action. Full closure requires its remaining causal and transmitting contributions, followed by a joint field/metric/scale solution and independent matter and cosmological predictions.

1. One hypothesis and a continuous physical test

Postulate. Spaces form inside spaces through collapse, localization and renewed expansion. A room can change its configuration and locally measured scales while inheriting the same dimensionless law. The central requirement is

$$\mathcal{T}_{\Theta}^{*}\mathbb{D}_{\Theta}=\mathbb{D}_{\Theta}.$$

Its block form has room operators on the diagonal and parent/child boundary maps off diagonal. The proposed invariant partition and present spectral prescription are

$$Z_{\Lambda}^{\mathrm{one}}(\mathbb{D}_{\Theta})=\int \mathcal{D}\varphi\, Z_{\Lambda}(e^{-\varphi/2}\mathbb{D}_{\Theta}e^{-\varphi/2})$$

$$S_{\mathrm{one}}=\mathrm{Tr}\, e^{-\mathbb{D}_{\Theta}^{2}/\Lambda^{2}}+\langle J\Psi,\mathbb{D}_{\Theta}\Psi\rangle.$$

The displayed functional is a formulation to complete, not an already normalized quantum-gravity path integral. The regulator, normalization scale, compensator, measure, determinant phase, zero modes and physical domain must be derived consistently. Induced geometric terms must be counted once. Adding a separate gravitational weight to rescue a stationary root would change the adopted construction.

The same action must generate the source of its geometry, the physical particle sectors and their interactions, and the observable response. The operational chain is:

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common action + state + recursive boundary domain
       -> boundary response and absolute stress
       -> coupled geometry and state evolution
       -> matter, energy transfer and physical modes
       -> expansion, clustering and lensing
       -> independent observational tests
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2. Six explanations the programme aims to establish

Each row is a research target. The third column records the calculation that would turn its interpretation into a physical result.

Target	Strongest present connection	Required physical closure
Constants and thermal limits	Characteristic/current calculations; compact-to-light matching; a flat thermal conversion control of the finite proper-time factor	Complete physical couplings, common causal cone and physical thermal state
Finite black-hole continuation	Regular benchmark geometry; horizon transport and a parent-matched canonical massless Dirac source with negative neck null	Self-sourced collapse, physical stability and complete continuation
Dark matter and dark energy	Boundary self-energy, parent-matched source residuals and energy-transfer identities	Joint background stress and perturbation predictions
Particle, wave and measurement	Field excitations, a projected Dirac–gauge interaction and a finite Gaussian state functional	Derived interactions, detector records and quantum statistics
Mass, chirality and antimatter	Exact invariant sheet/chirality sector, compact Dirac masses and charge algebra	Actual boundary coupling, physical sector selection and gauge representations
Probability and recursive infinity	Tail endpoint criteria; dilution of homogeneous free state differences; a specified determinant remainder	Consistent recursive state measure, causal duration and observable limits

2.1 Constants, resolution and absolute zero

Open prediction. The causal cone must follow from the physical principal symbol and agree across the relevant propagating species. A numerical value of a dimensional speed also depends on clock and ruler conventions. Derive the gravitational response before identifying

$$\ell_P = \sqrt{\hbar G/c^3}$$

with a scale of the model. A spectral cutoff does not by itself prove a shortest measurable distance; that requires a localized-probe/backreaction experiment. The historical constant dictionary names intended outputs but does not derive nature's values (constant dictionary).

Open prediction. Zero temperature is a limit of the physical many-body thermal state, not the bottom of the signed first-quantized Dirac spectrum. A suitable equilibrium construction must be independent of the arbitrary energy origin, handle ground-state degeneracy and zero modes, and distinguish residual spin or vacuum correlations from thermal excitation. The historical phrase equating absolute zero with a spectral infimum is therefore a proposed dictionary entry requiring this additional derivation.

2.2 Black holes and historical horizons

Imported result. Regular black-universe solutions already combine a trapped region, positive minimum areal radius and expanding interior in one geometry [6]. The project uses that solution as a development benchmark. Its existence is not proof that our universe is such an interior or that the same geometry is sourced by this operator.

Imported result. Maldacena, Milekhin and Popov already derive a semiclassical Einstein–Maxwell throat supported by charged massless Dirac vacuum energy [32]. Their sections 2 and 5 give the light magnetic channels, the anomaly-corrected Casimir stress and its Einstein matching. This is imported theory, not an NSC-computed solution, an observed wormhole, a generic formation process, or a child-expansion proof. The earlier radial transport operator does not contain their magnetic flux state. The charged-sector record checks compact parity, anomaly cancellation and heat-coefficient conventions; the later flat-holonomy calculation adds a conditional gauge saddle on a closed cell. The magnetic return geometry, mass, internal representation, vacuum/gravitational/gauge coefficients and global boundary matching remain unresolved. Their optical or redshift length is not proper length. Short-distance formulas have stated conditions, and longer separation or mouth stabilization require their sections 5.5–6. The retained ultraviolet volume term is not set to zero to obtain their asymptotically flat exterior.

Repository derivation. The actual horizon-penetrating radial Hamiltonian includes the shift and spin connection. Conserved Dirac norm transfers in the child direction in the declared Cauchy problem (tetrad, transport). The trapped finite boundary has an outflow structure; it cannot be replaced by an arbitrary reflecting wall.

Repository derivation. Section 4.8 records a phase-resolved parent-matched canonical massless Dirac candidate on that same geometry. Its neck null contractions are negative, but density, pressure anisotropy and flux do not match the required two-derivative tensor. The candidate is therefore not a self-sourced continuation.

Open prediction. Complete physical continuation requires a dynamically sourced solution, constraints, finite tidal response and extension of relevant finite-affine endpoints. Infinite coordinate time is not affine completeness (clock and horizon audit). High-energy Euclidean spectral form factors [13] are not by themselves a Lorentzian theorem that bosonic signals stop at a physical wall.

2.3 Dark response from adjacent rooms

Postulate. Degrees of freedom outside a locally resolved room can act through its boundary response. **Repository derivation.** Eliminating those degrees of freedom gives a Schur self-energy. This establishes an exact response mechanism, not its identification with the observed dark-matter and dark-energy components.

An isotropic background source and a clustering/lensing response are different projections of the same required effective stress. Both must be calculated. A finite vacuum cutoff does not solve the cosmological-constant problem without a predicted renormalized curvature and physical scale. Nor does a nonzero boundary map establish a continuing energy supply.

2.4 Localized excitations, waves and measurement

Imported result. Individual detections build interference patterns [1]. QFT describes particles as field excitations; solitons and collective coordinates supply additional useful construction patterns [4]. The existing particle/wave record is an imported collective-coordinate reduction. It does not solve an electron or detector.

Open prediction. Couple an identified excitation to a detector through the derived interaction and evolve the whole energy/charge ledger. Interference, localized records, outcome probabilities, Bell correlations and no-signalling must follow from that model. Decoherence or mechanical resonance alone does not derive the Born rule or select an individual outcome. Bell and PBR results constrain admissible accounts [2,3]; unresolved room labels alone do not establish one.

2.5 Chiral mass and charge-conjugate states

The scalar sheet benchmark in Section 3 provides an exact version of the proposed left/right relation. Its coupling has not yet been derived from the throat. Chirality, radial component, angular sector, sheet label and charge are distinct operations. Antimatter requires the conjugate gauge representation, not merely an opposite sheet or a reversed spatial gradient.

With the stated gamma conventions, distinguish the charge-conjugation matrix $C=i\gamma^2\gamma^0$ in $\psi^c=C\bar{\psi}^T$ from the antilinear column-spinor map $\psi^c=i\gamma^2\psi^*$. The quantum symmetry operation and a c-number conjugation map are not interchangeable definitions. Both opposite-parity compact projectors remain; sheet exchange is not charge conjugation. Physical photons require a spin-one gauge sector. The produced Dirac pairs do not establish baryonic dust or the observed matter excess.

2.6 Probability, infinite generation and existence

Interpretive hypothesis. Infinity describes ongoing generation rather than one static container. **Repository derivation.** On a particular unweighted spatial chain, the endpoint is uniquely determined for $\Omega\ q\le 1$, while $\Omega\ q>1$ requires data at infinite depth (tail result). This is an endpoint criterion, not a universal strict-contraction theorem; critical recursions can converge algebraically.

A nonterminating decimal is not evidence of nested geometry. Infinite room count is not proof of infinite mapped proper time. A physical claim of eternal generation requires the state measure, clock map, causal continuation and limiting observable to be defined. Energy conservation and a finite cutoff alone do not prove that child formation is inevitable or that existence has probability one.

3. A precise shared mass and self-energy benchmark

Repository derivation. In sheet-first Weyl order, consider

$$H_8 = I_2 \otimes \boldsymbol{\alpha}\cdot\mathbf{p} + \Phi\tau_1 \otimes \beta$$
$$\Pi_- = (I_8 - \tau_3 \otimes \gamma^5)/2.$$

The projector selects parent-left plus child-right. Exact calculation gives

$$[H_8, \Pi_-] = 0$$
$$W^\dagger H_8 W = \boldsymbol{\alpha}\cdot\mathbf{p} + \Phi\beta$$

where W embeds those four components. Thus one invariant sector is an ordinary massive Dirac Hamiltonian, with

$$E^2 = |\mathbf{p}|^2 + \Phi^2$$
$$\Sigma_p(E) = \Phi^2(E + \boldsymbol{\alpha}\cdot\mathbf{p})^{-1}.$$

The complementary sector is equally invariant and spectrally equivalent. Physical selection must follow from the common action or domain, not preference. A spinor-identity sheet link instead has energies ±p±Phi and is gapless at p=Phi. A Hermitian pseudoscalar link has the same free gap and self-energy as the scalar benchmark. The spectrum therefore does not determine the complete interaction. The observable bridge checks these distinctions and the corresponding charge algebra.

Both chiral components carry the same Dirac charge. A massive positive-energy eigenstate has a constant chirality expectation. The familiar left-to-right oscillation belongs to a specified initially chiral state containing both energy signs. In the full paired angular continuum problem, the massless radial geometry and its radius pulse preserve physical chirality. Their radial gap is consequently not already a four-dimensional chirality-breaking rest mass.

The next test is the actual boundary map's Clifford structure, orientation and invariant domain, followed by a controlled local limit and physical pole. These calculations are kinematic connections within this project; ordinary Dirac algebra is prior physics.

Repository derivation: evaluated boundary channels. The full paired-angular calculation now evaluates an energy-dependent four-component spatial link in an explicit current-conserving domain. Direct joined, transfer and Schur calculations agree below 2e-15 relative; an independent continuum integration gives 0.00947%–0.03645% errors at the reported resolution. The link preserves physical chirality. Its action-form Clifford decomposition has vector and axial content; scalar and pseudoscalar terms vanish to numerical precision (chiral boundary).

This result specifies the actual transport structure of that massless spatial realization. The proposed chiral-sector mass reduction requires an additional interaction or physical domain mechanism derived from the common action. A passive change of the child spin frame does not supply it: chirality must be transported with the frame. The numerical map is a finite spatial point-port response, with its units and regulator stated explicitly; it is not the full PG retarded map or a local mass term.

Imported result / repository derivation: compact mass identification. The missing source input is a physical four-dimensional mass, not another spatial gap. Applying the established conformal Dirac and Kaluza–Klein reductions [33,34] to the declared free carrier on an interval of coordinate length 2 $L_\star$, with the right-handed component vanishing at both endpoints, yields one Weyl zero mode and the massive Dirac tower

$$m_n = \frac{n\pi}{2L_\star}, \quad n \geq 1.$$

The full spacetime warp cancels from canonically normalized free evolution. This is an application of prior mathematics, recorded as an explanatory note rather than an extra compact result. Compact size, species, occupations and the actual throat coupling remain open (compact mass map).

Repository derivation: torsion interaction overlaps. Substituting those NSC profiles into the published higher-dimensional Einstein–Cartan contact [35] produces a mode-matrix vertex. After fermionic antisymmetrization the 3111 coupling remains nonzero, so a single massive level is not an exact interacting sector. The geometric warp changes the overlap coefficients even though it cancelled from free evolution. The five-dimensional stiffness s_T, compact boundary action, physical state and absolute stress remain inputs (compact interaction).

Imported result / repository derivation: charged parity and anomaly compatibility. To use the imported charged throat as a source test, a local realization must contain an actual gauge connection. The present candidate is a U(1) extension with two equally charged bulk copies; it does not derive electromagnetism or the internal algebra. Within this field content, opposite compact parities are required both for an even neutral scalar link and for cancellation of the zero-mode cubic and mixed gravitational–gauge anomalies. The two allowed projectors are the invariant scalar-link sectors identified above:

$$P_{\mathrm{even}} = \frac{I_8 \mp \tau_3 \otimes \gamma^5}{2}.$$

Both remain. Sheet swap is not charge conjugation: a right-handed charge-one field is counted as a left-handed charge-minus-one field only in anomaly bookkeeping (charged sector). The imported Einstein and Casimir solutions are not recomputed. Link mass, flux geometry, renormalized coefficients and child expansion remain unresolved.

The two five-dimensional Dirac copies yield one four-dimensional Dirac zero field before a link mass is added. Their ultraviolet spinor trace has rank eight; applying the bulk coefficients to this candidate requires that two-copy multiplicity, not the one-copy normalization of the earlier UV control.

4. Geometry, vacuum stress and an explicit energy ledger

Repository derivation. The isolated radial throat is gapless. Periodic confinement can create a band gap, and a smooth periodic profile removes the earlier geometric seams:

$$r(x) = \sqrt{a^2 + [\sin(kx)/k]^2}$$
$$k = \pi/(2R).$$

At a=1 and R=2,4,8, its continuum first band edges are approximately 0.76423204, 0.51425546 and 0.28454085 (smooth geometry). R remains an input. These are spatial spectral values, not identified particle masses.

The same profile requires a negative effective Einstein null source,

$$8\pi G(\rho + p_x) = -2r''/r.$$

The covariant vacuum controls establish negative axial-null vacuum response on a specified compact constant cylinder. The varying-neck calculation differentiates all metric functions and computes a periodic-minus-antiperiodic stress difference. Its local finite counterterms cancel in that difference; its absolute stress is not thereby fixed. Choosing a spin structure because its stress has the desired sign would not derive it.

A finite term proportional to the squared Weyl curvature leaves the anomaly unchanged but changes the varying-neck shape equation. The current flat common-scale integral also diverges in its finite realization. The ultraviolet functional, complete compensator and state must resolve these questions before a stationary root represents self-sourcing.

Repository derivation: finite metric response. The general-metric calculation now isolates four independent bulk channels in the local basis $M^4, M^2R, C^2, R^2, E_4, \Box R$. On the closed smooth cell, the Euler and divergence terms have exact zero bulk variation. Both R=2 and R=4 profiles distinguish the remaining four coefficients; the constant-cylinder control has rank two. Independent lapse, radial metric and radius variations agree with direct energy changes (finite terms).

For the declared finite-action sign convention, the exact neck null response is

$$\Delta(\rho + p_x)_{\text{fin}} = \frac{4M^2}{a^2}c_R + \frac{32(1+a^2k^2)}{3a^4}c_C - \frac{16(1+4a^2k^2)}{a^4}c_{R^2}.$$

This gives a concrete ultraviolet matching problem: determine these coefficients from the same functional, then solve the independent metric equations. They have not been fitted to make the neck stationary. The result concerns a closed torsionless metric sector through four derivatives; boundary contributions and other fields require their corresponding terms.

Repository derivation: five-dimensional ultraviolet matching. The leading proper-time bulk coefficients of the candidate torsion contact reuse the dimension-independent heat formulae [29,36]. In five dimensions the induced Einstein and K^2 terms have stiffness ratio 9. A matching window keeps this induced piece separate from the complementary one-loop determinant. The volume term is retained; it is not set to zero to manufacture an asymptotically flat solution. Leading stiffness 9 does not fix the finite datum c_T-$9c_R$. The displayed derivative scale is a coefficient comparison, not a physical torsion mass (UV map).

Imported result / repository derivation: quantum-flow limits. A published generalized Palatini flow [37] has a larger connection field space. Its metric contribution does not preserve the constrained Dirac three-form subspace in $d=5$; its fermion contribution does. Copying that paper's four-dimensional fixed-point numbers does not determine the present five-dimensional functional. The next flow must be defined on the actual metric, skew three-form, physical fermions, links and measure (flow compatibility).

Repository derivation: covariant determinant source. The static Euclidean operator on the same smooth cell is

$$D_\omega = \beta\omega/N + i\beta H_0(q,r).$$

Physical four-dimensional chirality anticommutes with D_ω. Its square retains the lapse gradient $\alpha\ \omega N'/(qN^2)$. The cutoff modulus per coordinate time is

$$E_\Lambda = \frac{1}{2}\sum_{\kappa\geq 1} 2\kappa\int_{\mathbb{R}}\frac{d\omega}{2\pi}\mathrm{Tr}_{X,4}\,E_1(D_\omega^2/\Lambda^2).$$

Frequency integration reproduces the known cylinder vacuum response. Independent metric variations conserve the static stress and recover the established Dirac heat coefficients $a_0=4$, $a_2=-R_E/3$ through a_4 [29]. Those coefficients are not a Nested-Space discovery. The specified ultraviolet subtraction is

$$E_{\mathrm{sub}}(M) = E_\Lambda - \frac{A_0\Lambda^4/2 + A_2\Lambda^2 + A_4\log(\Lambda^2/M^2)}{32\pi^2}.$$

At $a=1$, $R=2$ and $M=1$, the remainder energy is about -0.09050 with neck null stress about -0.00911 at $\Lambda=4$. This is a calculated negative null contribution under that subtraction, not a scheme-independent complete source (covariant source). On the same lapse/radius/shape probes the determinant now supplies actual projections $b_A^{\mathit{determinant}}$ in

$$b_A^{\mathrm{determinant}} + J_{Ai}c_i + b_A^{\mathrm{compensator,\,link,\,state}} = 0.$$

No coefficient is fitted to hold the neck open. Uniform clock scaling multiplies the covariant energy, while a fixed spatial-Hamiltonian cutoff does not.

Repository derivation: normalization profiles. Write $x=d^2/\Lambda^2$. The calculated modulus per eigenvalue is half the exponential integral E1(x). A smooth weighted logarithm and the heat-covariantized old rank term obey

$$g_{\log} = -\tfrac{1}{2}e^{-x}\log(d^2/M^2)$$
$$g_{\log} = g_{\mathrm{PT}} + [\log(M/\Lambda) + \gamma_E/2]e^{-x} + h(x)$$
$$h(x) = -\tfrac{1}{2}[e^{-x}(\log x + \gamma_E) + E_1(x)].$$

The remainder is not a multiple of the identity or of the heat kernel, so replacing the old finite-rank term by a heat trace changes the finite action. The identity is a comparison of prescriptions, not an insertion into Γ_{one} (normalization). Andrianov's primary projector $P_N=\Theta(1-D^2/\Lambda^2)$ is a different object [11]; its cylinder cutoff wall has a closed form, while an unrestricted $Tr1$ integral diverges.

Repository derivation. Assign half of the cross-region link energy to each region, with $h_A=(P_AH+HP_A)/2$, and use $C_{ij}=\langle c_j^\dagger c_i\rangle$. Then

$$\frac{dE_A}{dt} = \mathrm{Tr}(C\,i[H, h_A]) + \mathrm{Tr}(C\dot{h}_A).$$

For the static vacuum, $[H,C]=0$, so continuing net regional injection is zero. A prepared excitation transports conserved energy through the same geometry-derived link (energy transfer). Its current depends on occupations and correlations as well as the retarded map.

A prescribed smooth radius pulse gives exactly $H(t)=H_0+\varepsilon\, g(t)V$. It creates Dirac particle–antiparticle excitations. Independent evolution and spectral transition calculations agree, and supplied geometric work accounts for the excitation energy (vacuum work). At epsilon=0.02 and tau=0.5, one representative kappa=1 channel produces approximately 1.40181e-5 pairs and energy 3.46640e-5 in throat units. These are one-channel finite-amplitude controls, not cosmological rates. A uniform clock rescaling produces no pairs.

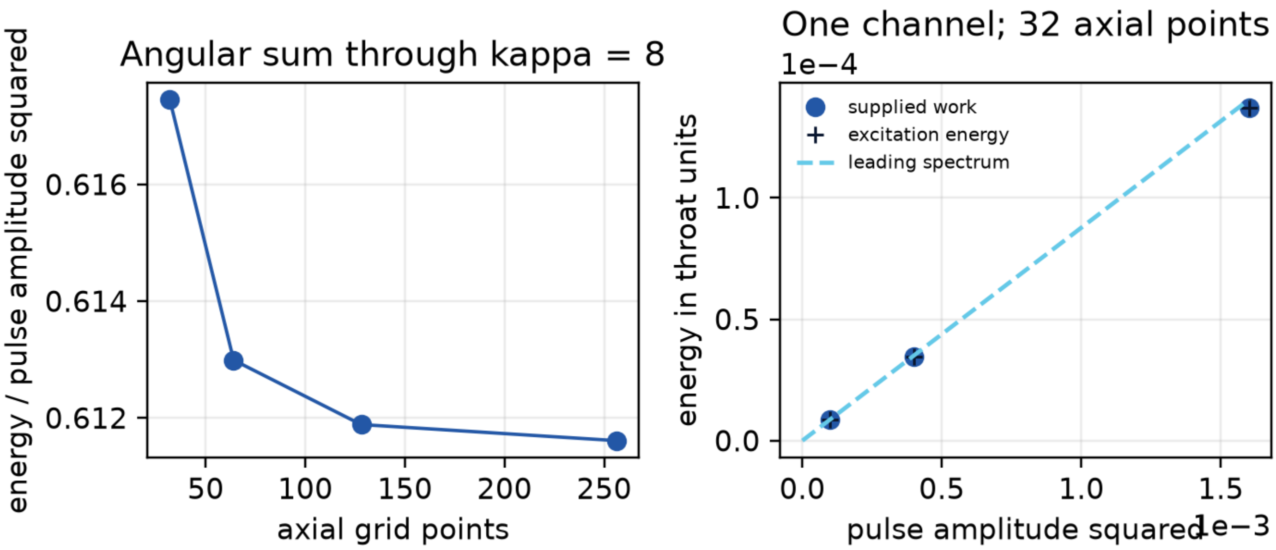

Figure. Response to a prescribed geometry pulse. Left: spatial refinement of the angular sum. Right: independent finite evolution accounts for supplied work in one channel. The pulse is an input; this is not a self-sourced cosmology. Vacuum-work evidence.

This is a calculated conversion from supplied geometric work into Dirac excitations. The deformation's own source and backreaction remain to be solved. A null-sign match, pair creation or a conserved finite energy ledger alone is not the complete metric equation.

Repository derivation: normalized histories and geometric noise. The same covariant operator supplies the canonical Hamiltonian $H=N^{1/2}H_0\,N^{1/2}$. For a Gaussian fermionic covariance $0\le C\le I$ and two history unitaries, the normalized closed-time-path amplitude is

$$Z[+,-;C] = \det(I - C + CU_-^{\dagger}U_+)$$
$$\Gamma_{\mathrm{IF}} = -i\log Z.$$

The Fock-space trace identity is due to Klich [30]; the present record implements it for this geometry vertex and checks mixed and pure states against an explicit eight-dimensional Fock calculation. Equal histories give $Z=1$. Causal contour derivatives reproduce the retarded commutator response, and the symmetrized noise matrix is positive. For the Gaussian time window already used in the pulse calculation, the smeared vertex variance equals the leading pair-production coefficient:

$$\mathrm{Var}(V_f) = N^{(2)}_{\mathrm{pairs}}/\varepsilon^2.$$

At $\tau=0.5$ this common weight is *0.03533459211* on the refined spatial grids (influence). Bare mean forces and equal-time fluctuations do not converge in the same way; they remain finite-regulator quantities, not renormalized absolute stress. Multiplying by a real local-action phase preserves equal-history normalization while changing the mean source, so $Z=1$ does not select the physical finite terms. The influence approach to gravitational response has prior work by Martín and Verdaguer [31]. The contribution here is the executable connection to this smooth Dirac geometry and its previous energy/pair calculation.

Repository derivation: matching static and causal response. The radius channel now connects the Euclidean frequency operator to the Lorentzian Hamiltonian response. Independent matrix integration of the ordinary finite Dirac loop equals the spectral vacuum susceptibility at imaginary external frequency, to about 10^{-15} on the tested finite grids. The finite proper-time Hessian is calculated separately, with its regulator retained.

The periodic-minus-antiperiodic difference removes state-independent local metric terms on identical geometry. It permits the continuum matching control

$$K_E^{P-AP}(\nu) \longrightarrow \chi_R^P(i\nu) - \chi_R^{AP}(i\nu).$$

The independent retarded sum uses a positive Abel transition weight and a stated cutoff extrapolation. These finite regulators are distinct. Through angular label $\kappa=8$, the two methods give:

Imaginary frequency	Euclidean relative radius response	Retarded extrapolation minus Euclidean
0	0.185771550	-1.49e-9
0.6	0.172525449	-1.41e-9
1.4	0.130625412	-1.14e-9

Spatial, angular, internal-frequency and cutoff refinements are recorded separately (response matching). The omitted angular tower has no proved remainder bound. A change from inverse-radius to log-radius coordinates adds the required contact $\langle H'' \rangle$; its relative value is nonzero and is retained. With that term, independent metric differentiation agrees with the earlier covariant source Hessian. This is a verified matter-response connection in the smooth ultrastatic radius channel. Absolute finite coefficients, finite-cutoff causal completion and constrained metric backreaction remain to be determined.

4.1 One source must satisfy the geometric boundary equation

Repository derivation using established boundary calculus. The compact chiral domain fixes the leading Einstein boundary term of the determinant magnitude. It is isotropic for the Lorentzian Hamiltonian current, but the Euclidean Dirac operator requires complementary adjoint data. Using $D_E^\dagger D_E$, the known mixed heat coefficients [29] yield the Einstein/Gibbons–Hawking–York ratio 2, no leading boundary-volume term, and a specified geometric a_3 contribution (compact boundary action).

This is a coefficient and domain match, not a boundary held in place by its own source. The Gaussian compact endpoints have nonzero geometric momentum. The common metric variation therefore requires

$$\pi_p^{ab} + \pi_c^{ab} + \frac{\delta\Gamma_{\mathrm{rest}}}{\delta h_{ab}} = 0.$$

Here Γ_{rest} excludes the geometric terms already included in π_p,π_c. The variation uses established gravitational boundary calculus [41]. Reflecting compact endpoints and a transmitting parent–child throat are distinct domains.

4.2 A finite Dirac interaction on the curved cell

Imported formulation / repository calculation. The published massless interval determinant [38] can be applied to the existing covariant four-dimensional operator. For two Dirac copies and the untwisted transverse interval of conformal length $\ell=2L_\star$, its finite endpoint interaction is

$$\Gamma_{\rm int}=-\tfrac{1}{2}\mathrm{Tr}\log\left(1-e^{-2\ell\sqrt{D_4^2}}\right).$$

The half-rank zero level and full massive-level multiplicities are retained. Differentiating this function of the same D_4 supplies independent lapse, radial-metric, radius and separation responses. No scalar stress source is inserted. These are effective four-dimensional sources integrated over the transverse interval, rather than a pointwise five-dimensional stress or a tension localized at one endpoint.

For the unit-neck ultrastatic cell with axial circumference 4 and transverse length 2, the compact interaction gives

$$(\rho+p_r)_{\rm int}=-8.9162\times10^{-4}.$$

The units belong to the recorded g_4 geometry. Matrix variation agrees with independent energy differences; an independent Bessel-frequency integral and separate spatial/angular/frequency refinements agree. The local conservation residual decreases with resolution (source and state record).

The first two local curvature terms of this finite interaction are derived and kept in the same energy account. A decompactified bulk determinant is generally nonlocal in curved g_4; it cannot be discarded using a scaleless flat-space subtraction. Sections 4.5–4.6 now provide the full free compact cutoff weight and its light-field matching for the declared reflecting domain. The common completion, state and transmitting phase remain necessary for the physical metric equation.

4.3 A conditional state-selection mechanism

The null-source sign depends on global geometry and state. With the same transverse length and unit neck, enlarging the axial circle to 8 changes the interaction's neck null source to $+1.6402\times10^{-4}$. Periodic axial data can also reverse the sign. These changes are physical differences between prescribed finite cells, not an error bar proving the isolated-throat limit.

Where the charged ultrastatic geometry contains a closed axial return path, a flat U(1) connection has a physical Wilson line. Its effective spatial phase combines the spin structure and gauge holonomy:

$$\alpha=\eta_{\rm spin}-\tfrac{1}{2\pi}\oint A_x\,dx.$$

The phase is defined modulo one. Local gauge-invariant terms do not distinguish these flat connections. The general boundary determinant [39,40], applied to the positive radial partner operators, makes each free-fermion contribution minimal at effective AP phase, $\alpha_\star=1/2$. The full free-KK determinant independently gives an AP-minus-P energy of about -0.16995535 and positive second variation 3.19343937 in the recorded cell. This is a one-loop minimum of the real potential when the holonomy is free to vary, not a derivation of the complete quantum state or physical mode spectrum.

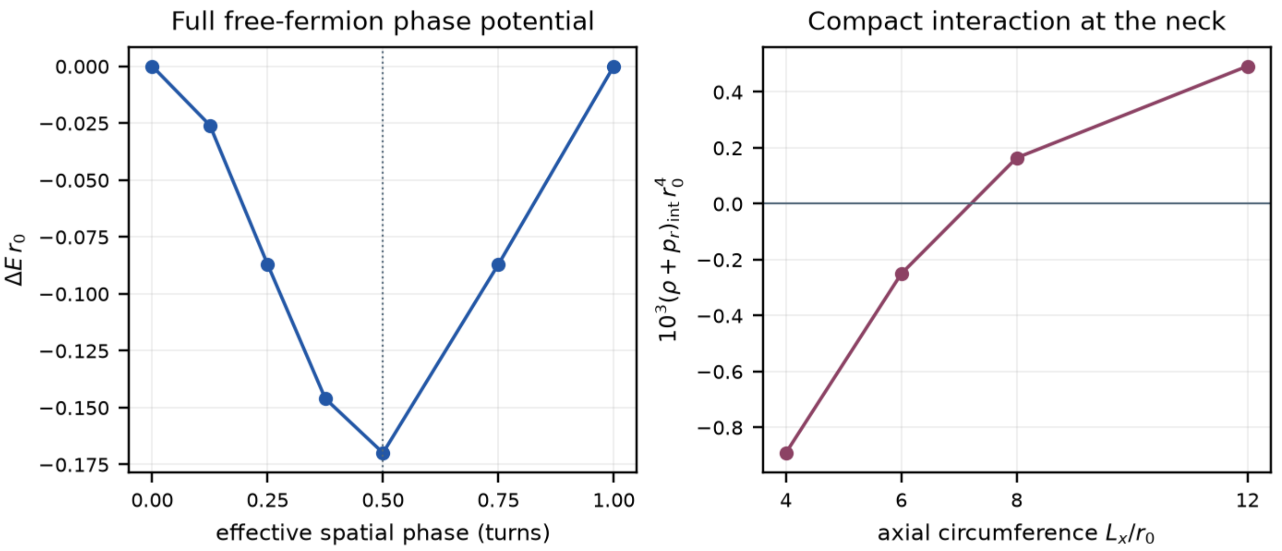

Figure. Left: full free-KK holonomy energy differences on the ultrastatic closed cell; the effective AP phase minimizes this one-loop real potential. Right: only the compact endpoint interaction's effective null source, at that phase. The transverse separation is 2 in g4 neck units. Increasing the axial circle changes the global geometry and the source sign. The remaining bulk source and transmitting geometry are still required. Source and state record.

The Casimir and Wilson-line mechanisms are established physics [38–40]. The project calculation matches them to this operator and identifies their source, normalization and domain dependence. The recursive geometry must supply the actual return path; an unwrapped chain acquires no Wilson loop merely because the development cell has one.

4.4 Apply the source to the unwrapped horizon

Repository application of established Dirac and conformal-state results. The unwrapped transition patch has spatial form R times S2 times the transverse interval. It has no non-contractible radial cycle supplying the finite cell's flat Wilson line. A declared magnetic flux instead supplies known angular zero modes: one four-dimensional Dirac field gives $|q_{mag}|$ complex two-dimensional fields [32,43]. The integer flux is not selected by this calculation.

For this free massless sector, Unruh boundary data and the stored horizon surface gravity give the parent Killing power [44]

$$P_\infty = \frac{|q_{\rm mag}|\kappa_h^2}{48\pi}.$$

At $\kappa_h=0.2383257996$, this is 3.76661306×10^{-4} per unit $|q_{mag}|$, in the recorded natural units. The local horizon limit is regular under the specified state conditions. This power is not yet a child proper-volume density rate or the cosmological Q. Interactions, a possible light-field mass and the global state remain additional conditions (horizon/source record).

The result tests an actual source requirement. The sector's horizon null stress is negative, but both null contractions at the old imposed neck are positive, where that geometry requires negative ones. Static magnetic and potential terms have zero radial null contraction. The minimal sector therefore does not source that prescribed neck by itself. The reduced metric equations keep the missing source U(r) explicit and propagate their constraint; they do not choose U from the desired geometry. A nonzero outgoing flux also requires evolution or compensating flux before it can belong to an exactly static solution.

4.5 The compact quantum warp supplies a spectral weight

The classical conformal reduction preserves the canonical compact mass tower. Its quantum determinant has a different normalization. For the conformal metric

$$g_{5,s} = e^{2s\sigma(Y)}(g_4 + dY^2)$$

transport to fixed L2 measure gives

$$D_s = e^{-s\sigma/2} D_0 e^{-s\sigma/2}$$
$$L_s = D_s^{\dagger} D_s.$$

The compact chiral values and complementary adjoint domain are retained. In the canonical field, the eigenproblem uses both weights, $\int e^{-s\sigma}|D_0\chi|^2=\varepsilon\int e^{s\sigma}|\chi|^2$. Its compact eigenvalues define

$$h_{\Lambda,s}(y) = \sum_j E_1\big(\varepsilon_j(y;s)/\Lambda^2\big)$$
$$\Gamma_{5,\Lambda} = \tfrac{1}{2}\mathrm{Tr}_4\, h_{\Lambda,s}(D_4^2).$$

The two complementary five-dimensional copies are already included. This is a finite-cutoff free Euclidean determinant magnitude, with a separate prescription required for actual spacetime zero modes. It composes with the existing D4 rather than replacing it. The physical compact warp is calculated at s=1; the numerical homogeneous application uses $g_4=R^2\times S^2(r)$.

For $q_{mag}=1$, r=1, transverse length 2 and cutoff 2, the potential per reference two-dimensional area changes from 2.218882505 to 2.154980560. This is a 2.88% change in that regulated contribution. Independent first variations and a dual-heat conformal identity agree within the stated Galerkin convergence. The warp effect does not imply a changed classical mass or a cosmological fraction (warped-source record).

4.6 One coefficient account for the retained light field

Separate one canonical light Dirac field at a matching cutoff nu by defining

$$H_\nu(y) = h_\Lambda(y) - E_1(y/\nu^2)$$
$$\Gamma_{5,\Lambda} = \Gamma_{\mathrm{light},\nu} + \Gamma_{H,\nu}.$$

The compact proper length fixes the finite limit $H_\nu(0)$. Two remaining spectral moments determine the vacuum and Einstein terms. Applying the established four-dimensional heat/Mellin formulas [29,45] gives the complementary Euclidean density

$$\mathcal{L}_{H,E} = V_D - A_D R_E + C_{F,D} F^2 + \cdots.$$

At cutoff 2, transverse length 2, s=1 and matching cutoff $\nu=1$:

Dirac contribution	Coefficient	Units
Vacuum V_D	0.1725408633	inverse length to the fourth power
Einstein A_D	0.00550228445	inverse length squared
Gauge $C_{F,D}$	0.00483725070	dimensionless

The curvature-squared terms are fixed by the same $H_\nu(0)$. The Dirac a4 combination contains no independent R-squared term; that does not eliminate an undetermined finite term of the completed theory. The source map retains the established static curvature-sign convention. Its local truncation needs small external curvature, derivatives and gauge field; that approximation has not been established at the imposed neck.

Changing nu transfers terms between the light field and its complement:

$$\partial_\nu \Gamma_{H,\nu} = -\partial_\nu \Gamma_{\text{light},\nu}.$$

The numerical vacuum, Einstein and gauge changes cancel. These coefficients are therefore linked contributions in a stated prescription; nu is not a knob for fitting a physical coupling or geometry. The remaining common functional and causal state must still be matched before identifying the complete Newton constant, gauge coupling or U(r) (compact/light matching).

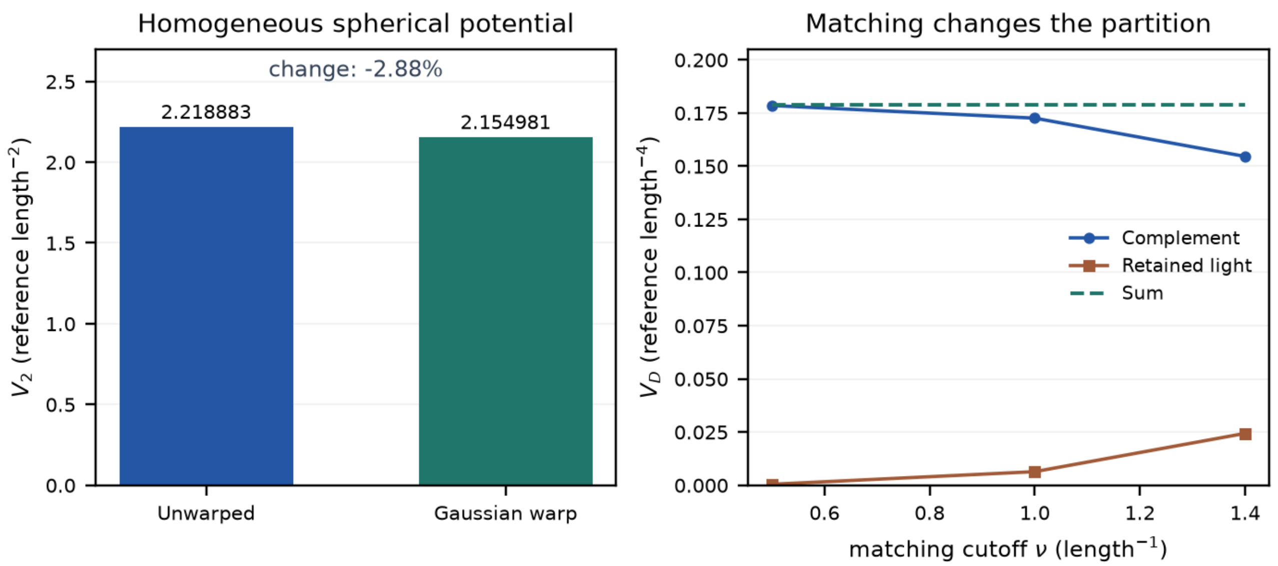

Figure. Left: two-dimensional-area potential on R2 times the unit sphere, with magnetic flux label 1. Right: the four-dimensional local vacuum coefficient split between a retained light field and its complement; their sum is unchanged. Both use cutoff 2 and transverse length 2. The panels have different dimensions and are different projections of the same free determinant. They do not plot cosmological evolution. Warped source; light-field matching.

The source equation now has a defined interface: combine the computed free determinant with the contribution required by the common measure, compensator, transmitting interaction and state. These records test distinct controlled settings; their assembly into one Lorentzian solution remains part of physical closure. In particular, the spherical Einstein term already present inside the determinant must be counted once. No additional independently weighted gravitational action is introduced.

The next subsections follow that interface rather than nine separate programmes. A definite interaction among already retained light fields, a first-order spherical geometry owner, and a local curvature correction fix the low-energy source language. At the stored child curvature the local truncation is not controlled, so the original spectral weight is used. Homogeneous free-state differences then dilute, a canonical massless reference is distinguished from a parent-matched candidate, and the finite proper-time factor is shown to be state-dependent. Published methods are reused; none of the following calculations claims a new general theorem or a fitted Newton constant.

4.7 A retained interaction, a spherical owner and a curvature contact

Imported formulation / repository application. The retained Dirac and spherical-photon fields already supply a projected interaction in the magnetic lowest-Landau-level sector. Established two-dimensional bosonization [56] reduces the charged current to one collective mode with screening scale $\mu^2(r)=|q|\, g_4^{\,2}/(4\pi^2\, r^2)$. That mode is not an inserted fundamental scalar, a resolution compensator or a predicted four-dimensional rest mass. The physical matched g_4 is not selected here, and the Dirac complement coefficient $C_{F,D}$ is not its complete inverse coupling.

Outside the recorded horizon the collective potential is nonnegative and vanishes at both scattering ends, so a constant-mass Boltzmann factor is the wrong flux calculation. With the specified Unruh data and no incoming parent bath, the projected Killing power is the free central-charge term plus the charged Bose integral of the transmission. Reference [44] supplies the free flux normalization; the charged transmission integral is evaluated in the gauge-source record. At the stored $\kappa_h=0.238325799634$, for $q=1$ and unselected test couplings, $g_4=0.5$ gives parent Killing power 0.0003631182, or 0.964044 of one free LLL channel. These are synthetic coupling controls, not a fit of the physical coupling or a cosmological dark fraction. Stationary current conservation and future-regular Painlevé–Gullstrand components convert that power into a horizon null source $T(K,K)_h=-P\_\infty/(4\pi\, r_h^{\,2})$. A nonzero net power still requires evolution or compensating flux before it can belong to an exactly static geometry (gauge-source record).

Imported formulation / repository application. The local two-derivative spherical sector of the same action now has an explicit first-order owner. Spherical reduction of $S_4=\int\sqrt{|g_4|}[-A_4\, R_L-V_4-C_F\, F^2]$ with dimensionless area $X=8\pi\, A_4\, r^2$ recovers the Grumiller–Kummer–Vassilevich first-order theory [46], including the area-dependent charge potential. The auxiliary torsion of that first-order patch is not the independent five-dimensional skew torsion of the earlier Einstein–Cartan candidate. Coupling the existing charge mode closes the classical constraint algebra, including the $1/X$ area force that is not the factorized matter potential printed in that reference. This identifies an action-specific canonical measure and ghost operator for the reduced theory. It replaces a guessed one-dimensional scale weight; it does not quantize the full nonlocal spectral action, select physical couplings, or evolve a new geometry (spherical-action record).

Imported formulation / repository application. The known local curvature-squared contribution can be retained in that same spherical system as a first-order effective-field-theory contact. A checked metric redefinition [47,48] converts the bulk C^2 and R^2 terms into

$$\Delta\mathcal{L}_m = -\frac{c_W}{2A^2}\left(T_{\mu\nu}T^{\mu\nu} - \frac{T^2}{3}\right) - \frac{c_R}{4A^2}T^2$$

without assigning extra initial data to a fourth-order truncation. The charge/Maxwell source, including its angular pressure, supplies a definite spherical contact. The first-order constraints still close at this EFT order. A change of variables does not cancel physical vacuum curvature: pulling the map back to the original metric removes the apparent cosmological-constant shift. The construction requires $|c_i\, R|/A$ and $|c_i\, T|/A^2$ small. That regime is not established at the imposed neck, and the nonlocal remainder $h(D_4^{\,2})$ still belongs to the theory. This is not a replacement of the spectral action or a proof of physical-mode health (curvature-EFT record).

Imported spectrum / repository composition. The stored development geometry reaches curvatures at which that local expansion is not controlled. At the recorded horizon, neck and asymptotic child, the largest sectional magnitudes divided by $\nu^2=1$ are 0.2168, 4.7124 and 9.4248. Composing the existing compact weight h_Λ with the four-sphere Dirac spectrum [49] therefore supplies the usable Euclidean endpoint response. At the child-radius control $a=0.325735$, the full logarithmic-radius derivative is 0.001239600, while the local truncation through a_4 gives -0.164273768. A local stationary radius near *0.43742* has full derivative *0.084326*, so it is not stationary in the same determinant. Near agreement at larger a supports the expected low-curvature use of the local coefficients. The round sphere is not the global continuation of the black-universe chart, whose interior has Kantowski–Sachs spatial topology. Use the original spectral functional at these curvatures; do not accept the local root as a solution of Γ_5 (spectral-endpoint record).

4.8 Canonical reference, parent matching and the neck budget

Three objects must be kept distinct: a renormalized massless reference, a parent-matched state on the actual domain, and the finite-Λ full action. Asymptotic tail formulae are not substituted at the neck.

Repository derivation using established Dirac stress differences. For the fixed free compact tower on the actual expanding interior, a defined class of homogeneous state differences has vanishing stress density. Trace-norm conservation and Hölder's inequality give

$$|\Delta\rho| \le \frac{M + K/r}{4\pi s_{\min} r^3}$$

with $O(r^{-4})$ for massless modes ($M=0$) and an $O(r^{-3})$ envelope when finite massive occupation moments are present. Expansion reduces the local stress through redshift and proper-volume growth; it does not erase the state. The free LLL Unruh-minus-Hartle–Hawking difference decays as r^{-4} in the interior, with asymptotic r^4 density $-\kappa_h^2/(576\pi^3)$. This is not a global Hadamard theorem, a cosmic no-hair result, or an identification of the absolute vacuum. A vanishing free difference does not imply that acceleration ends: a persistent common vacuum stress remains possible (child-state record).

Imported Hadamard/conformal construction / repository application. A massless reference can be specified on the child tail by smoothly continuing the conformal metric $\tilde{g}=g/r^2$ to an auxiliary ultrastatic region and propagating its ground covariance back [26,52,53]. In the inherited proper-time convention the leading absolute stress on that tail is

$$\lim \langle T^\mu{}_\nu\rangle_{\text{ref}} = \frac{11 H_{\text{child}}^4}{960\pi^2}\delta^\mu{}_\nu$$
$$H_{\text{child}} = \sqrt{3\pi}$$

Expanding that same conformal source gives the leading radial null combination $-u^2/16+O(u^3)$. That expansion is not valid at the neck, where u is not small. Converting the reference into the adopted finite cutoff requires

$$\Gamma_5 = \Gamma_{\text{light, ren}} + (\Gamma_{H,\nu} + \mathcal{C}_{\nu,\mu}).$$

The pieces depend on normalization; their sum preserves the original functional. Adding the renormalized reference to an unconverted complement would change Γ_5. This conversion is not a vacuum-tuning mechanism (massless-reference record).

Repository derivation. The same reference is then evaluated at finite child radius, retaining the full angular tower, covariant subtraction, conformal anomaly transport and the energy/work identity. Units remain $\hbar=c=L_{throat}=1$ and $\mu=1$. At the neck the auxiliary reference is $\rho=0.7101684$, $p_\parallel=0.5026775$, $p_\perp=0.0376558$, hence $\rho+p_\parallel=1.2128459$. After extracting one healthy Einstein term $a_{EH}=A_{EH}L_{throat}{}^2>0$, the imposed neck requires $L_{throat}{}^4\ T_{total,kk}=4a_{EH}(1-3\pi/2)=-14.8495559\ a_{EH}<0$. The calculated reference is positive and cannot supply that null requirement by itself. A pure vacuum term proportional to $g_{\mu\nu}$ cannot change the null component. No reference width was varied to obtain a stress sign (angular-stress record).

Imported characteristic-state prescription / repository calculation. Combining the standard fermionic Unruh boundary data [50] with the complex reflection amplitude of the actual exterior Dirac operator produces a definite massless source candidate on the expanding child. Affine-horizon occupation and empty incoming parent modes are compressed with the unitary scattering matrix; the factor i in the real interior spin frame is required by the known zero-frequency Rindler projector. The state was specified by those horizon and parent data before the neck sign was computed. In the same canonical heat convention the $N=32$ tail-corrected neck values are:

Component	Value in the declared throat units
ρ	-0.0033068962
$p_{\parallel}$	-0.0494945966
$p_{\perp}$	-0.0429957713
radial null +	-0.0528075900
radial null -	-0.0527953956
outward parent Killing power	0.0001422206795

The stress entries have dimensions L^{-4}; Killing power has dimensions L^{-2} with $\hbar=c=1$. Angular refinement from $N=16$ to 32, after the sixth-order tail correction, changes the tensor by less than 4×10^{-6} across the recorded radii. Frequency, quadrature and time refinements are assessed separately. These are measured sensitivities with an asymptotic angular-tail approximation, not a rigorous error bound on the infinite tower.

Both radial null contractions are negative after including the flux. The two-derivative geometry with $a_{EH}>0$ instead requires

$$(\rho, p_{\parallel}, p_{\perp}, T_{\hat{T}\hat{Z}}) = (2a_{\mathrm{EH}},\ 2a_{\mathrm{EH}}(1-3\pi),\ 2a_{\mathrm{EH}}(1-3\pi),\ 0).$$

The computed density has the opposite sign to that positive-Einstein requirement, the pressures are unequal, and the state carries outward parent Killing power 0.0001422206795. Matching only the null equation by choosing a_{EH} would conceal those residuals. Leave a_{EH} symbolic. The remaining common-action terms must supply the componentwise difference, or the geometry must evolve under the same action. This is not a global NSC Hadamard theorem, a complete finite-cutoff stress, or a cosmological Q: deposition, proper volume, clock matching and a solved geometry are still required (parent-matched source).

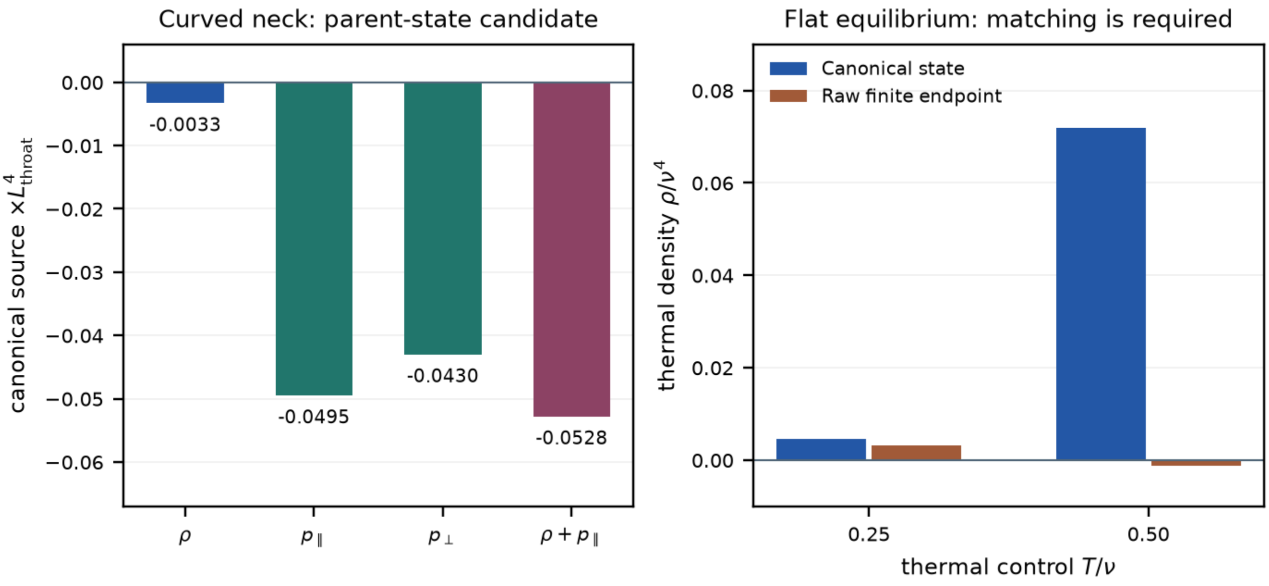

Figure. Left: the canonical massless Dirac source at the imposed curved neck. The last bar is the mean of the two radial null contractions; nonzero flux splits them slightly, with both remaining negative. Density, anisotropy and flux leave independent geometric residuals. Right: a separate flat thermal control shows the finite-endpoint/state conversion requirement. The panels use distinct geometries and normalizations. They do not depict cosmic evolution or assign a temperature to the child. Parent source; state/regulator control.

4.9 Finite-cutoff conversion is state dependent

Imported thermal determinant / repository compatibility check. The adopted finite Euclidean proper-time factor cannot be identified, without a state-dependent conversion, with the thermal source of the canonical Dirac field used above. On flat R^3 times an antiperiodic Euclidean thermal circle, with one ordinary massless Dirac field, $T/\nu=0.5$ gives canonical $\rho=0.0719658654\ \nu^4$ versus raw finite $\rho=-0.00122667356\ \nu^4$ after subtracting that prescription's own zero-temperature energy [51]. Local vacuum coefficients cancel in the thermal-minus-vacuum difference and cannot repair it. A pole-residue argument is insufficient: the regulated inverse has residue one on shell, yet the finite Matsubara determinant differs from the canonical CAR excitation energy. This is a conversion control, not a local temperature on the spacelike interior Killing generator, or a cosmological energy-transfer prediction.

If the intended physical field retains its canonical state functional, the completion must satisfy $\Delta\Gamma_{completion}=\Delta\Gamma_{canonical}-\Delta\Gamma_{raw}$ for that sector and common contour. The previously computed negative neck null remains valid in its canonical-massless scope; it is not yet the complete finite-cutoff source. The next owner is that same-state conversion, including massive, boundary and recursive sectors (state-regulator record).

5. Recursion and cosmological predictions

At a common dimensional energy, parent-normalized first-order elimination gives

$$\Gamma_p(x) = K_p(x) - \tfrac{1}{\Omega} b\,\Gamma_c(x/\Omega)^{-1} b^{\dagger}\,.$$

Here $b=B_{dim}/\Lambda_p$. The inheritance scale $\zeta=(\Lambda\ L_*)^2$ and cutoff ratio $\Omega=\Lambda_c/\Lambda_p$ remain distinct. Neither is selected by unit dilation alone. Finite chains and spatial endpoint estimates are verified; the covariant state, physical recursive domain and stationary scale remain open.

Let Q_b denote internal transfer into the visible component and J_b,J_d denote external room supply. The complete ledger is

$$\dot{\rho}_b + 3H(\rho_b + p_b) = Q_b + J_b$$
$$\dot{\rho}_d + 3H(\rho_d + p_d) = -Q_b + J_d$$

$$\dot{\rho} + 3H(\rho + p) = J_b + J_d.$$

Internal Q_b cancels. The Bianchi identity requires the full source of ordinary Einstein gravity to be conserved; external room supply therefore needs its reservoir/boundary gravitational accounting. A throat surface integral is power, and a worldtube integral is energy. A cosmological density rate additionally requires proper time, proper volume and deposition into the identified component. The parent-matched Killing power *0.0001422206795* is therefore not a cosmological Q.

For the closed flat constant-G baseline with visible dust, a constant fraction f below unity requires

$$\frac{Q_b}{H\rho_b} = 1 - 2q_{\mathrm{dec}}.$$

Zero Q_b alone does not force deceleration: dust plus positive Lambda has zero exchange and approaches de Sitter expansion. A finite regular plateau is an additional condition. A known scalar Q_b also leaves pressure and perturbations undetermined (plateau conditions, source requirements).

Predicting H(z), BAO and CMB requires the background and thermal history. Predicting clustering and lensing also requires pressure perturbations, anisotropic stress, momentum transfer and constrained metric response. Lambda-CDM is a predictive comparison model; CPL is a phenomenological equation-of-state parameterization. Neither is replaced merely by naming an unevaluated Q.

The frozen development targets are documented with their model assumptions in observational targets. H0DN reports 73.50±0.81 km/s/Mpc, with significance depending on the comparison [23]. DES Y6 reports S8=0.789±0.012, a 2.6-sigma projected difference from its primary-CMB comparison [24]. KiDS-Legacy reports S8=0.815 with +0.016/-0.021 uncertainty, agreeing with Planck at 0.73 sigma [25]. These are development targets, not NSC predictions or evidence that a boundary mechanism has resolved a tension.

6. Nuclear physics and the remaining common-action equations

The Skyrme/BPS sector is a possible low-energy construction pattern. Its coefficients must be obtained from the evolved common effective action before its particles, binding energies or stellar limits are predictions of NSC.

The often-quoted 3.34 solar-mass maximum is imported from a nuclear-calibrated Einstein–BPS model [5]. The historical runner records the published value rather than independently recomputing it. Its potential and calibrated nuclear coefficients matter; other potentials yield different maxima. It is not a universal collapse threshold or an unfitted NSC result (stellar benchmark).

The next coupled calculation must determine the same-state finite-cutoff causal conversion and complementary functional, together with massive, boundary and recursive sectors, then the independent metric, field and relative-scale equations. The parent-matched canonical massless tensor is a reusable input, not the complete source. The remaining finite terms are not fixed by equal-history normalization, by fitting G to a null residual, or by treating the flat thermal control as a child temperature. The same completed solution must then supply its constrained physical response before particle identities, nuclear interactions, collapse and cosmological predictions can be established with one parameter set. Further raw Dirac coefficient scans cannot determine a different functional contribution.

The joint closure is a candidate project contribution, not an established novelty claim. Primary ingredients retain their original attribution. Failure of one local realization informs its revision; it neither disproves the nested architecture nor confirms it.

7. Reproduction and scope

This working revision contains 100 records: 58 frozen historical records and 42 scoped follow-ups. The nine new source and state records, with their laboratory notes, are imported from laboratory commit *161028d*. All 91 earlier manifest step objects and scientific records/generators retain their bytes and comparison policies. Native schemas are preserved; publication identities and routing are supplied by adapters. Immutable source notes retain their creation-time publication status; this release manifest supplies their present publication state.

The new records compare their published fields, formulae, scope statements and source/input hashes under the absolute and relative tolerances stored in those JSON files. Earlier horizon-source, warped-source and compact/light policies are unchanged. For the spectral endpoint's small finite-difference residual, portable mode additionally checks the Richardson and subtraction identities, retains the generator's original derivative-accuracy requirement, and propagates the raw operands' comparison budgets. The raw derivatives retain their original tolerances; exact mode is unchanged. Structure, source/comparator hashes and scope fields remain exact. These reproduction policies are distinct from the convergence studies and physical assumptions in each record.

Use the default demonstration to inspect fourteen authenticated stored cases. It launches no scientific generators. Explicit recomputation and the full release gate are documented in reproducing. One integrated gate authenticates the dependency graph, runs relevant tests, regenerates the collection in isolation, and checks deterministic PDF output. Same-environment byte identity and portable all-field agreement are reported separately.

The following appendices preserve the main normalization corrections and finite mathematical controls. Their historical numerical roots remain diagnostics. Full covariant physical closure, predicted particle species, a measured dark fraction, and our universe's parent/child identification remain open.

Appendix A. Historical scale tests and corrections

The original numerical candidates are retained for provenance. This appendix does not define the current complete action or imply that adding a recursive tail alone supplies the missing finite measure and state.

A.1 Historical scale candidates

The inheritance scale is

$$\zeta = (\Lambda L_\star)^2$$
$$\mu^2 = 1 - \frac{3\pi}{2\zeta}.$$

Repository derivation. Matching the local on-room vacuum-form component *108/1015* to the doubled gap, then using the exact expanding child’s Ricci sign, selected a *conditional* one-scale gap $|\Phi|^2/\Lambda^2=1006/1015$ and a boundary cross-projection $\Xi_{boundary}=54/503$ (nsc-1-s-one-child-orientation.json). **Repository derivation.** The exact child asymptotic rejects that magnitude. The actual child vacuum is $\Lambda_{e,child} L_\star^2=18\pi$, so a real positive gap requires $\zeta>3\pi/2$. Direct $\Lambda L_\star=1$ gives a negative μ^2 and is not the physical child-matched branch (nsc-1-s-one-child-scale-correction.json). The identities for Φ as gap, Schur self-energy, and metric link remain. The numbers *1006/1015* and *54/503* are **superseded** as child-matched predictions.

The ZETA1 chain then searched for a child-allowed stationary scale of the joined-minus-disconnected Dirac spectrum on the global slice. Every root in the following table is a **numerical diagnostic**. None is a physical ζ. The later anomaly compensation supersedes all of them as stationary points of the one equation.

Compact record	What was computed	Diagnostic value	Why it is not physical ζ
nsc-2-zeta1-lowest-mode.json	Isolated lowest-mode heat minimum	$\zeta=5.096657218448003$, $\mu=0.27458356021684416$	Lowest sector only; heat trace, not the determinant; **superseded candidate**.
nsc-2-zeta1-angular-tower.json	Angular tower through $n=12$	Diagnostic $n=1$ heat root 5.096622100098284; $n=2$ removes that heat-only root; $n\leq 12$ minimum derivative 7.350807275212887	Heat-only truncation; **superseded candidate**; absence of a heat root is not a theorem against stationarity.
nsc-2-zeta1-regulated-determinant.json	Exponentially regulated joined-minus-disconnected determinant	$\zeta_{det}=4.99072409354754$, $\mu_{det}=0.2361577587050819$	Uncompensated determinant; warp, lapse, shift, and anomaly omitted; **superseded candidate**.
nsc-2-zeta1-y-boundary-sensitivity.json	Factorized compact-y controls	Periodic root 4.8765555133228515; Neumann root 4.79786026064662; Dirichlet and MIT bag: no root	Factorized unwarped control; a zero mode was not selected in order to force a root; **superseded candidate**.
nsc-2-zeta1-orbifold-parity.json	Joint fermion/metric S^1/Z_2 parity	$\zeta_{orb}=4.747863009733466$, $\mu_{orb}=0.08643829078229333$	Unwarped factorization; **superseded candidate**.
nsc-2-zeta1-warped-y.json	Exact conformal compact warp	$\zeta=4.748389947082489$, $\mu=0.08707307719677547$	Factorized radial-plus-y sum, not the full Dirac operator with lapse and shift; **superseded candidate**.

Every corresponding JSON record states that the candidate is not promoted, that lapse and shift or the local matrix anomaly remain omitted, and that the electron mass is not predicted.

A.2 Anomaly and inherited subtraction

Repository derivation. The invariant partition function requires cancellation of pure cutoff rescaling while retaining the physical change of the child-constrained gap. For one eigenvalue the physical child-link derivative after compensation is negative whenever the gap constraint $\mu^2=1-3\pi/(2\zeta)$ is imposed (nsc-2-zeta1-anomaly-decomposition.json).

At the warped determinant candidate $\zeta=4.748389947082489$:

Piece	Value
Determinant derivative	0.01201522080638507
Pure cutoff anomaly	12.336590431250533
Physical child-link derivative	-12.324575210444092

The compensated child-link derivative stays negative on the 257-point scan $3\pi/2<\zeta\leq 200$, from -19.422735759304963 to -0.5875504421475721.

Those historical numbers used the inconsistent conversion $(\lambda_j+\mu^2)/\zeta$. They remain numerical diagnostics of that proxy.

Therefore those determinant-only ζ values are invalidated as stationary points of the one equation. ζ_{det} and $\zeta_{warped\ y}$ are uncompensated determinant diagnostics, not inheritance-scale predictions.

Historical interpretation. The next missing contribution was identified as the recursive child tail, rather than an independently weighted Einstein–Gauss–Bonnet or heat action (nsc-2-zeta1-anomaly-owner-correction.json). Geometry is already the compensating anomaly of the same fermionic determinant. Adding that geometry again is forbidden double-counting. The calculations through the scan joined one parent to one child treated as a terminal asymptotic domain. Recursion requires that the child's boundary response already contain its own child tail. The compact `nonclaims` include `recursive_tail_is_known_to_restore_a_scale_root: false`.

Related routes already rejected on their own JSON nonclaims, and not repaired by retuning Φ or adding a second geometric weight:

- $\Lambda L_\star=1$ as the physical child-matched gap;
- the lowest-mode heat-only scale minimum;
- independently weighted Einstein–Gauss–Bonnet or heat geometry as the missing scale owner;
- pure critical five-dimensional Einstein–Gauss–Bonnet as the complete interface law;
- the standalone bosonic heat-trace graviton covariance as a positive Stieltjes propagator ($g''(0)=228/175>0$; nsc-1-s-one-reflection-positivity-obstruction.json);
- the inverse relative heat Hessian as Osterwalder–Schrader positive metric covariance (the full 31-time reflection matrix has minimum eigenvalue -1.3679466754324912; the source record counts fourteen negative modes while portable Linux counts thirteen because one near-zero mode crosses the numerical counting tolerance; nsc-1-s-one-relative-reflection-test.json);
- constant two-wall shadow matter as the dominant kiloparsec gravitating mass.

The fermionic sheet-resolution observable is Osterwalder–Schrader positive in the controlled flat sector (nsc-1-s-one-fermionic-relative-observable.json). That does not make σ_3 the complete graviton, and it does not restore a scale root. Full-exponential positivity in that sector is a heat-kernel diagnostic, not the complete physical anomaly.

A.3 Corrected units and the sign theorem

Repository derivation. The ZETA1 definitions are $\lambda_j=L_\star^2 eig_j(D_{spatial}^2)$, $\zeta=\Lambda^2 L_\star^2$, and $\mu^2=|\Phi|^2/\Lambda^2=1-a/\zeta$ with $a=3\pi/2$. The same declared additive-gap operator therefore requires

$$q_j = \frac{\lambda_j}{\zeta} + \mu^2 = 1 + \frac{\lambda_j - a}{\zeta}.$$

The historical runners implemented $(\lambda_j+\mu^2)/\zeta$, a physical mass squared smaller by $1/\zeta$ than the declared mass squared. Compact record nsc-2-zeta1-unit-closure-check.json reconstructs the old derivative as a control, then changes only that conversion. It retains radius 30, 750 radial half-intervals, angular sectors 1–12, 32 compact intervals, and the historical factorization and angular weights.

At the old warped candidate $\zeta=4.748389947082489$:

Quantity	Historical argument	Argument with consistent units
Determinant logarithmic derivative	0.01201522080638507	-24.319952223294422
Derivative after inherited cutoff subtraction	-12.324575210444092	-36.58292832809548

The independent old-argument reconstruction differs from the stored derivative by 1.53×10^{-8}. With radial spacings 0.08, 0.04, and 0.02, the corrected subtracted derivative is -36.58813190, -36.58292833, and -36.58162543. Its sign is resolved.

Repository derivation. For each mode the determinant piece is one half of $E_1(q_j)$. At fixed spatial geometry, subtracting the same pure-cutoff term used historically gives

$$\frac{d\Gamma_{\mathrm{sub},j}}{d\log\zeta} = -\frac{e^{-q_j}}{2q_j}.$$

The disconnected radial matrix is the principal submatrix obtained by deleting the throat node. Hermitian eigenvalue interlacing and the positive decreasing weight $f(\lambda)=e^{-q(\lambda)}/q(\lambda)$ then imply that the total subtracted derivative is strictly negative for every $\zeta>3\pi/2$ in this finite, frozen-geometry, additive-gap family. The *257*-point scan corroborates the sign; it is not the basis for extending the statement between sample points. A diagnostic root of the unsubtracted finite determinant near $\zeta=6.09675392014$ still has subtracted derivative about -14.15469, so it does not solve the adopted subtracted scale equation. No physical scale is inferred.

Consequence. Refining this family or selecting another determinant-only minimum cannot close its scale equation. A further attempt needs a derived change in the physical operator, its geometry/link dependence on scale, or the anomaly prescription. Merely naming the omitted recursive tail does not prove that it supplies that change.

Repository derivation. For the same relative-sheet symbol

```
D_δ =
[ p e^(-δ)   Φ         ]
[ Φ          -p e^(δ) ]
```

the unregulated finite determinant is independent of δ. The proper-time functional used later in ZETA1 has a strictly positive regulated relative Hessian at $\delta=0$ for $p\neq 0$. Raw determinant invariance therefore cannot justify setting the regulated relative Hessian to zero. The primary anomaly paper distinguishes its normalization scale from the spectral cutoff (arXiv:1106.3263).

Appendix B. Finite operator and boundary controls

B.1 Finite regulated variations and a spacetime obstruction

Repository derivation. For a Hermitian finite Dirac matrix with no unresolved zeros, one explicit proper-time prescription is (nsc-3-regulated-recursion.json; docs/nsc-regulated-recursion.md)

$$\Gamma_{\Lambda,M}(D)=\tfrac{1}{2}\mathrm{Tr}\,E_1(D^2/\Lambda^2)+N[\log(M/\Lambda)+\gamma_E/2].$$

Its first and second variations retain noncommuting perturbations through divided differences. A finite scale cocycle $C(t)=\Gamma_{reg}(D_t)-\Gamma_{reg}(D)-t\mathrm{Tr}\Sigma$ is derived rather than discarded; its endpoint matches an independent integral of $C'(t)$. Normalization M and cutoff Λ remain distinct. This finite orthonormal-basis calculus is not a derived continuum gravitational measure.

Repository derivation. An ultrastatic Euclidean product $D_E^2=-\partial_\tau^2+H^2$ produces a frequency factor $\mathrm{Tr}_{sp}e^{-tH^2}/\sqrt{4\pi t}$. That factor changes the variational functional; a spatial heat trace cannot silently inherit a four-spacetime anomaly.

Imported result / repository witness. The recorded horizon-penetrating coordinate-time Hamiltonian is not elliptic in the trapped region: at the throat $\beta^2=3\pi/2>1$, and a nonzero spatial covector makes one principal eigenvalue exactly zero. Elliptic spatial heat calculus therefore cannot be imported as the full spacetime determinant. The general loss of ellipticity at horizons is documented by Finster and Röken (arXiv:1512.00761); the project contribution is the explicit witness for this carrier, not discovery of the phenomenon.

Along $D(t)=e^{-t}D$ at fixed cutoff, normalization, and finite domain, $d\Gamma/dt=\mathrm{Tr}e^{-D(t)^2/\Lambda^2}>0$. No isolated common-scale extremum exists in that finite family. Normalized finite recursion with $1/\Omega$ agrees with direct chain inversion in declared-matrix controls. H and b there are not yet computed throat maps. The compact `nonclaims` include `full_covariant_action_derived: false` and `physical_stationarity_solved: false`.

B.2 The isolated radial throat is gapless

Repository derivation. The unwarped spatial radial Dirac operator already declared by the project,

$$D = -i\sigma_2\partial_\rho + \sigma_1\frac{\kappa}{\sqrt{1+\rho^2}}$$

has essential spectrum equal to R (nsc-3-radial-spectrum.json; docs/nsc-radial-spectrum.md). A constructive Weyl sequence of compactly supported H^1 bumps escaping to infinity shows that every real energy belongs to the essential spectrum. Smooth radial geometry and current-conserving gluing alone do not generate a nonzero spectral mass gap. Finite-box levels are not physical masses. There is no global L^2 zero mode. The result does not cover the compact warped, interacting, recursive, or Lorentzian operator, and it does not reject Nested-Space Cosmology.

B.3 A controlled geometric gap from repeated throat geometry

Repository derivation. Repeating the finite segment $\rho\in[-R,R]$ with $w=(1+\rho^2)^{-1/2}$, instead of two asymptotically widening ends, produces an open spectral gap without an inserted constant mass (nsc-3-geometric-chain.json; docs/nsc-geometric-chain.md). At zero energy the exact motif transfer multipliers are $\exp(\pm\, 2\operatorname{asinh} R)$; neither has modulus one, so zero is absent from every real Bloch fibre. Continuum first band edges and second-order lattice agreement are

R	Continuum first band edge	Finest-lattice absolute error
2	*0.7034881641*	6.88×10^{-6}
4	*0.4552655377*	6.20×10^{-6}
8	*0.2479700183*	2.43×10^{-6}

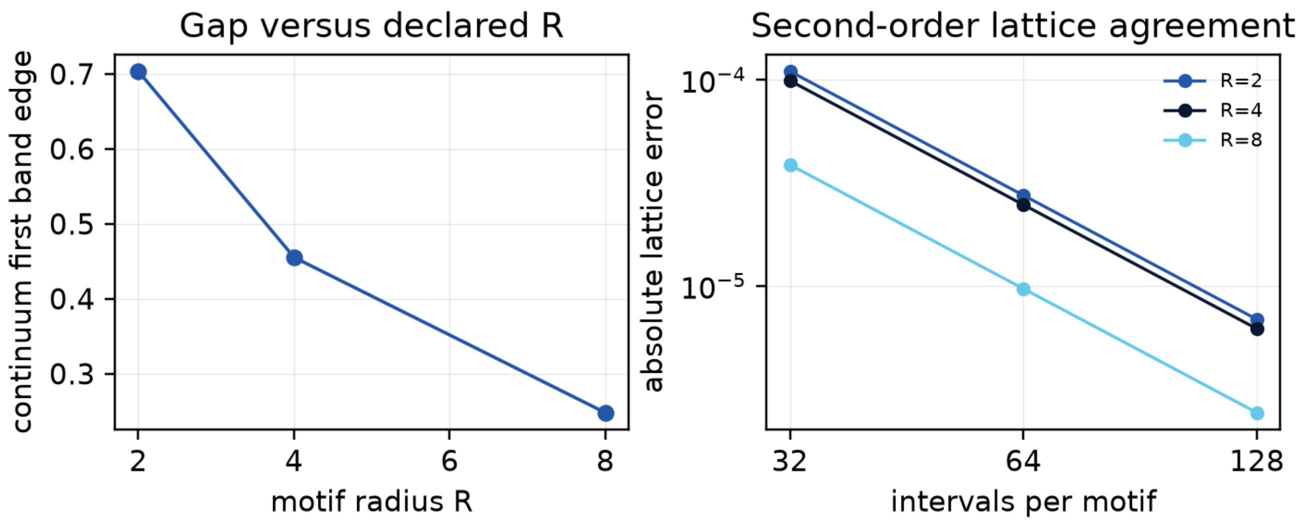

Figure. Continuum first band edges and lattice errors from nsc-3-geometric-chain.json. R is a declared input.

The next-room link is the staggered stencil entry $B_{last\ edge,first\ node}=1/h+w_{last}/2$, not a fitted mass. Normalized finite recursion with that geometric B agrees with direct multi-motif inversion. R and Ω remain declared inputs. Seams are periodic in w but gravitational junction equations are unproved. This is a positive project-level controlled realization of geometric confinement, not a new-to-world Floquet discovery and not a Lorentzian nested cosmology or an electron.

B.4 Computed finite boundary maps and the threshold distinction

Repository derivation. The finite first-order radial calculation now evaluates parent and child Weyl maps with sparse LU, throat reconstruction, independent ODE integration, and direct Schur controls (nsc-3-boundary-response.json). The parent lies at positive radial coordinate and its outward throat normal is negative; the child lies at negative radial coordinate and its outward throat normal is positive. The oriented relation is

$$N_p(E)+N_c(E)=E[m_c(E)-m_p(E)].$$

A nonzero real-energy probability-current probe tests the local conservation law. The compact warp calculation reports mode-mixing leakage: its average-warp expression is a projected approximation. These results compute finite spatial maps; the full horizon boundary maps, physical compensator, and Lorentzian geometry equations remain open.

Repository derivation. The same isolated radial geometry supplies an explicit threshold counterexample. Solving the upper squared-Dirac equation with unit throat value and vanishing exterior value gives, for t=asinh(R),

$$I_p=e^{3t}/6+e^{t}/2-2/3$$
$$I_c=2/3-e^{-t}/2-e^{-3t}/6$$

$$N_p+N_c=I_p^{-1}+I_c^{-1}\longrightarrow 3/2.$$

Independent ODE solutions reproduce these maps (nsc-3-threshold-response.json). The positive boundary stiffness coexists with the gapless radial bulk spectrum; it is not a particle mass of 3/2. A mass identification requires the full pole problem, residues and physical state. At zero energy m(E) can be singular, so E m(E) must be evaluated by its limit or by the squared-operator boundary problem.

Appendix C. Compact, UV and charged-throat controls

This appendix records technical controls behind the compact-sector and source matching. It does not add a compact-mass result record, recompute the imported wormhole, or select a physical coupling.

The compact-mass application uses four-dimensional signature +--- and a fifth gamma matrix $i\gamma^5$, giving five-dimensional signature +----. It applies Fischmann's conformal law and the interval reduction of Grossman and Neubert. The zero mode is a single Weyl field; it cannot be passed to a massive four-component source as though it contained both chiralities. Opposite endpoint chirality supplies the mirror zero mode and the same massive tower. Self-adjointness alone does not choose between them.

The compact-interaction overlaps are dimensionless profile integrals after the bulk Einstein normalization

$$\kappa^2_{4,\,\mathrm{bulk}}=\frac{\kappa_5^2}{I_3},\qquad I_3=\int e^{3\sigma}dY.$$

For the declared warp, $I_3/L_\star$=1.96509101243726. The 3111 left-current fermionic component remains of order 6 after antisymmetrization; the 0111 vertex vanishes by the even bulk warp. These signs are not energy-density signs.

For one bulk Dirac field, the ultraviolet map retains

$$\frac{4(\Lambda^5-\nu^5)}{5(4\pi)^{5/2}}\int\sqrt{g}\,.$$

Omitting that volume term in an unconstrained metric variation would change the source problem. Finite coefficients c_R and c_T are left unspecified; leading cutoff compensation leaves c_T-9c_R unchanged.

For the imported throat, the short-exterior-length relations of Maldacena, Milekhin and Popov include

$$r_e^2=\frac{\pi q^2 G_N}{g^2},\quad E_{\text{quantum}}=-\frac{q}{8\ell}.$$

The approximation uses large flux, weak gauge coupling and specified exterior geometry. The optical/redshift scale ℓ is not the proper throat length. Exterior matching and mouth stabilization must satisfy the source's assumptions; these lengths are not substituted as NSC predictions.

C.1 Vacuum, gravity and gauge coefficients must match together

For the retained local action $A\ R\text{-}CF^2\text{-}V$, with positive A,C, define $G_N=(16\pi A)^{-1}$, $g^2=(4C)^{-1}$ and $\lambda_4=V/(2A)$. The magnetic charge-radius convention of [32] gives

$$\Xi=\lambda_4 r_Q^2=\frac{q_{\text{mag}}^2 VC}{8A^2}.$$

The integer magnetic flux q_{mag} is distinct from the radial-metric factor and cosmological deceleration parameter. The common five-dimensional positive spectral coefficients obey $\Xi_{vol}\geq q_{mag}^2$; the nonnegative proper-time class gives the stronger $\Xi_{vol}\geq 3q_{mag}^2$. This follows from their Mellin and warp-moment inequalities. In contrast, the standard positive-vacuum charged extremal seed requires $\Xi\leq 1/4$ [42]. The controlled Gaussian warp and compact Einstein boundary correction do not bridge that mismatch (coefficient condition).

This excludes using those retained coefficients alone as the near-extremal seed of [32]. It does not exclude the complete NSC functional or all charged geometries. A vacuum subtraction, another positive profile or an additional copy count is not selected to force that seed.

C.2 Compact source and holonomy scope

For the declared compact chiral projector, the Euclidean adjoint has complementary boundary values. The resulting mixed condition has $S=-K\ P/2$, with the inward derivative and outward mean-curvature convention stated in the boundary note. The leading boundary term supplies the required Einstein/GHY ratio; the actual Gaussian endpoints still require a source balance.

The curved compact interaction is a finite piece of the determinant. It does not include the decompactified bulk term, formally $-\ell Tr|D_4|/2$, or the conformal boundary cocycle. Its flat-space volume coefficient and induced Einstein coefficient are fixed by the known interval and heat formulas [29,38]. Subtracting those two terms from the full interaction is exact bookkeeping, not a short-curvature approximation at arbitrary $\ell^2|R|$.

For the holonomy result, an ultrastatic proper-radial coordinate gives $L_\pm=-\partial_s^2+W^2\pm W'$, $W=\kappa/r>0$. The possible zero solutions are not periodic. With $t=\omega^2+m_n^2$, the discriminant satisfies $\Delta(t)>2$. The positive determinant has phase dependence $\Delta(t)\text{-}2\cos(2\pi\alpha)$ [39,40]. Each fermionic $-c\log[\Delta\text{-}2\cos(2\pi\alpha)]$, $c>0$, decreases on $0<\alpha<1/2$ and has positive curvature at the AP minimum. This fixes the conditional one-loop real-potential saddle. It neither selects a temperature nor proves that the physical recursive domain contains this loop.

Appendix D. The source matching equations

D.1 State and horizon conditions

In the two-dimensional conformal frame $ds^2=A\ du\ dv$, the free sector obeys

$$T_{uu}=\frac{c_{\text{eff}}(2AA''-A'^2)}{192\pi}+t_u$$
$$T_{vv}=\frac{c_{\text{eff}}(2AA''-A'^2)}{192\pi}+t_v.$$

The mixed component is $T_{uv}=c_{eff}AA''/(96\pi)$. Here $c_{eff}=|q_{mag}|$ is the free field count. Unruh data use $t_u=c_{eff}\kappa_h^2/(48\pi)$, $t_v=0$; the Hartle–Hawking control has equal incoming and outgoing terms. The parent Killing current is t_u-t_v. Its spherical projection divides by the proper area $4\pi\ r^2$, and the recorded PG transformation controls the horizon limit. This fixes the source within the specified sector and state; it does not construct a global NSC vacuum [44]. The full source derivation also gives the reduced metric constraint and the original-neck comparison.

D.2 Keep the complex-spinor and measure contributions

For the compact pair, choose $D_0=\lambda\sigma_1$-$i\sigma_3\partial_Y$. A real matrix representation uses the phase basis $\chi=(f,ig)$, for which

$$|D_0\chi|^2 = (\lambda g - f')^2 + (\lambda f - g')^2.$$

The cross term integrates with the nonconstant weight $e^{-s\sigma}$; it cannot be removed by restricting both physical components to real values. The direct complex-field norm, matrix quadratic form and scalar quadrature agree in the warped-source check. The independent conformal variation retains both adjoint domains before using the paired-copy symmetry. Omitted-sector heat bounds are distinct from the observed convergence of retained Galerkin eigenvalues.

D.3 Light-field normalization and the remaining functional

Define the compact norm and proper length by

$$J = \frac{1}{\ell}\int e^{s\sigma(Y)}\,dY$$
$$\ell_{\text{proper}} = \ell J.$$

The light singular branch has $\varepsilon_0(y)=y/J^2+O(y^2)$, and

$$H_\nu(0) = \log\frac{\Lambda^2 J^2}{\nu^2} + 2\sum_{p\geq 1} E_1\Big[\Big(\frac{p\pi}{\ell_{\text{proper}}\Lambda}\Big)^2\Big].$$

The canonical light mass remains zero. Define the two moments

$$Q_1 = \int_0^\infty H(y)\,dy$$
$$Q_2 = \int_0^\infty yH(y)\,dy.$$

The established trace expansion [29,45] gives

$$V_D = \frac{2Q_2}{(4\pi)^2}$$
$$A_D = \frac{Q_1}{6(4\pi)^2}$$
$$C_{F,D} = \frac{H_\nu(0)}{3(4\pi)^2}.$$

For the old static energy basis, R_E=-R_L; squared invariants and the static box-R term agree. The matching record gives that conversion and the finite changes at three matching cutoffs. The normalization mass of a determinant, the matching cutoff nu, and the physical Lambda are different quantities.

The unresolved causal assembly is now written with the computed reference and state pieces explicit:

$$\Gamma^{\mathrm{CTP}}_{\mathrm{one}} = \Gamma^{\mathrm{CTP}}_{\mathrm{ref,\,ren}} + \left(\Gamma^{\mathrm{CTP}}_{H,\nu} + \mathcal{C}^{\mathrm{CTP}}_{\nu,\mu}\right) + \Delta\Gamma^{\mathrm{CTP}}_{\mathrm{state}} + \Gamma^{\mathrm{CTP}}_{\mathrm{remaining}}.$$

The last term denotes the contribution still required by the common measure, compensator, remaining interactions and boundary/recursive sectors. The free massive compact determinant is already included in Γ_H and is counted once. It is not an independently adjustable gravitational action. Adding the canonical parent-matched stress to an unconverted cutoff complement would double-count terms. The Euclidean matching identity does not determine the initial state, cross-correlations, heavy-sector causal continuation or transmitting phase. Those are the remaining inputs to the same-action metric equations, not invitations to retune the computed coefficients or to fit G.

Appendix E. Interaction, spectral endpoint and state conversion

This appendix records the technical controls behind Sections 4.7–4.9. It does not add a tenth source programme, recompute imported wormhole or thermal-field theorems, or select a physical coupling.

E.1 Projected gauge transmission

The charged collective mode is the canonical two-dimensional current/electric-displacement field. There is no four-dimensional minimally coupled r''/r term in its exterior potential. Angular pressure follows by varying r in the reduced action and must not be dropped. For this real stationary potential, scattering reciprocity equates horizon emission with the computed transmission. Independent scalar-field initial-value checks agree at three frequencies; relative current defects are below 6×10^{-12} in the three base coupling calculations. Boonserm–Visser tail bounds [55] enclose the $g_4=0.5$ charged-power ratio between about 0.9640376 and 0.9641120. Higher angular photons, Landau levels, compact excitations and magnetic catalysis remain outside the calculation.

E.2 Spherical constraints and the EFT contact

In the first-order variables of [46], with $V=U\,X_+\,X_-+V+W_m$, the three smeared classical brackets close after the area-dependent charge potential is included in the structure functions. The local temporal-gauge ghost operator has formal determinant $(\det\partial_0)^2\det(\partial_0+X_+\,U)$. Geometry/ghost cancellation in that patch does not remove the matter measure. At a positive-radius minimum the map $r\to X$ remains invertible even though $X'=0$; taking X itself as a spacetime coordinate fails where its gradient vanishes.

The curvature correction is first order in a bookkeeping parameter ε. Its constraint increments are $\Delta G_1=0$, $\Delta G_2=e^+{}_1\,F$, $\Delta G_3=-e^-{}_1\,F$ with $F=d(X)K_0{}^2+F_0(X,\chi)$ and $d(X)=-c_W/(2AX)$. Closure is not asserted for a resummed $G_0+\varepsilon\,G_1$ at arbitrary ε. Extra matter or neutral CFT sectors belong in the total $T_{\mu\nu}$ before it is squared.

E.3 Full spectral versus local radius derivatives

The two-copy compact weight on S^4 is

$$\Gamma_5(a) = \tfrac{2}{3}\sum_{m=2}^{\infty}(m^3-m)\,h_\Lambda(m^2/a^2).$$

There is no additional copy factor. The comparison retains the canonical light field exactly and approximates only the matched complement through a_4. On S^4 the Weyl tensor and $\Box R$ vanish; the integrated Euler term is constant.

Sphere radius a	**Full** $\partial_{\ln a}\Gamma_5$	**Local** $\partial_{\ln a}\Gamma_5$
0.325735, matching the stored child *H*	*0.001239600*	*-0.164273768*
0.43742254, local stationary control	*0.084326048*	approximately *0*
1	*14.8033344*	*14.8393135*
2	*284.848997*	*284.854519*
4	*4754.593055*	*4754.594317*

The isotropic action response at the child-radius control is $\Pi_a \approx 0.001045912$. It already includes induced geometric terms and must not be added as a second matter stress.

E.4 Reference conversion and angular subtraction

At the stored child curvature radius $a=0.325735$ and $\nu=1$, the invariant density projection splits as renormalized massless reference *0.103125000*, original finite-cutoff light contribution 2.8628×10^{-16}, converted complement *-0.102079088*, and unchanged total Dirac modulus *0.00104591235*.

The finite-radius angular calculation subtracts two-dimensional adiabatic orders *0*, *2* and *4* before the angular sum [54], then restores the four-dimensional proper-time finite part with harmonic coefficient $L_R=\ln(\mu R)+\gamma_E/2$. Smooth windows act on the already convergent remainder; they are not a physical regulator or a change of state. The auxiliary reference at five areal radii is recorded in the angular-stress JSON; only the neck value enters the source budget above.

E.5 Parent-matched covariance

In the exterior current basis, $H_{ext}=-i\sigma_3\partial_x+V_\kappa\sigma_1$ with $V_\kappa=\kappa\sqrt{A}/r$. For positive frequency, $f=(1+e^{2\pi\omega/\kappa_h})^{-1}$ and $s=\sqrt{f(1-f)}$. Compressing occupied affine-horizon data with the scattering matrix yields the interior covariance

```
C_(in)=
[ 1-f      i s R^* ]
[ -i s R   f|R|^2  ]
```

Its eigenvalues are *0* and $1-f|T|^2$. The charge-related momentum channel is $C(+\omega)=I-\sigma_3\, C(-\omega)^*\sigma_3$. The two channels give equal diagonal stress and a nonzero momentum flux. A development version that omitted the factor *i* was corrected before any accepted result. Global extension, spin/current trace maps and microlocal regularity remain explicit conditions of physical use; Gérard–Häfner–Wrochna [50] is a characteristic-state prescription on its stated domain, not an automatic theorem on the NSC black-universe extension.

E.6 Flat thermal conversion table

The control domain is flat R^3 times an antiperiodic Euclidean thermal circle. The cutoff ν is the light-sector matching cutoff, distinct from the carrier Λ and from temperature.

T/ν	**Canonical thermal** ρ/ν^4	**Finite-endpoint thermal** ρ/ν^4
0.10	*0.0001151454*	*0.0001151454*
0.25	*0.0044978666*	*0.0031350475*
0.50	*0.0719658654*	*-0.0012266736*
1.00	*1.1514538468*	*-0.0063244282*

At $T/\nu=0.5$ the record's finite-endpoint density is $-0.00122667356\ \nu^4$. The discrepancy is exponentially small at very small T/ν, but it is not an exact identity. A flat control at the stored parent Hawking temperature also retains the logarithm of its tiny mismatch; that number is not an estimate for the curved child source.

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